DOCGUARD FORENSIC AUDIT CERTIFICATE

CERTIFICATE ID: DG-37F101A7-31E2-49
ISSUED: 2026-10-30 19:46:09 UTC

Automated Multimodal Document Tampering & Cryptographic Integrity Verification

Standard: ISO/IEC 27037 & FRE 902(14)

FORENSIC VERDICT: AUTHENTIC / CLEAN VERIFIED

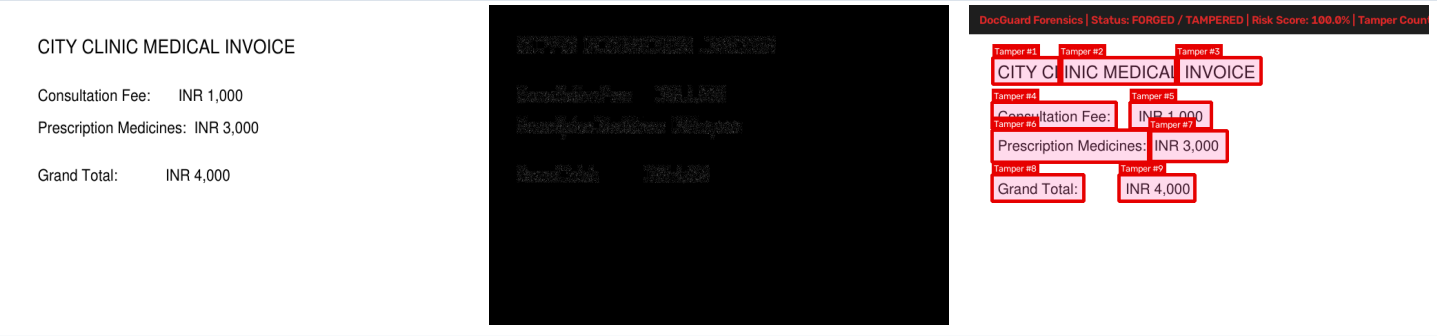
The analyzed document passed all multimodal neural, compression, noise residual, and structural forensic checks (Global Forgery Risk Score: 12.50%). No manipulation detected.

FORGERY RISK SCORE
12.5%
Threshold: 50.0%

1. Legal Chain of Custody & Document Identity

Original Filename:	**sample_invoice_pristine.pdf**	**File Size:**	1.61 KB (1,645 bytes)
Analysis Time:	2026-10-31 01:16:09 (Local)	**Audit System:**	DocGuard Automated Forensic System v1.0
SHA-256 Checksum:	**F2E8C73B9720AAC2857D24A9A3AD25B9CA469FBC2C55C375C104226E0D32DA34**	**MD5 Checksum:**	**2582A71FA38885A2CA8A0DB8D7255CC6**

2. Tri-Panel Visual Forensic Evidence



Panel A: Raw RGB Document
Normalized Document Canvas

Panel B: In-Memory ELA Map
JPEG Compression Residuals

Panel C: XAI Anomaly Map
Bounding Boxes & Translucent Overlay

3. Six-Factor Forensic Scorecard Breakdown

Forensic Vector / Factor	Methodology / Analysis Engine	Observed Metric / Diagnostic	Safety Threshold	Status
1. Deep Multimodal Fusion Brain	Late Fusion (ResNet-50 + EfficientNet-B0)	12.5% Probability	< 50.0% Risk	**PASS**
2. Error Level Analysis (ELA)	Differential JPEG Q=95 Quantization	Uniform Compression Equilibrium	Uniform Residuals	**PASS**
3. Spatial Noise Residuals	3x3 Median Spatial Residual Variance	Consistent High-Frequency Distribution	Homogeneous Poisson Noise	**PASS**
4. PDF Binary Architecture	Linearized %%EOF Revision Audit	Single Clean File Header / Trailer	Single Trailer (Count == 1)	**PASS**
5. Font Seams & Typography	Embedded Type1/TrueType Font Audit	Standard Homogeneous Font Descriptors	Consistent Kerning & Fonts	**PASS**
6. Deterministic Arithmetic Ledger	Symbolic Rule Verifier (sum(Items) == Total)	Sum(Items) == Grand Total (Exact Match)	Discrepancy == 0.00	**PASS**

4. Symbolic Arithmetic Ledger Consistency Audit

Extracted Line Items Sum:	INR 4,000.00	**Declared Grand Total:**	INR 4,000.00
Arithmetic Discrepancy:	**INR 0.00**	**Ledger Audit Verdict:**	**PASSED (BALANCED LEDGER)**
Detailed Ledger Finding:	*ARITHMETIC VERIFIED: Sum of 2 line items (4000.00 INR) matches declared Grand Total (4000.00 INR) with zero discrepancy.*		

5. Forensic Adjudication Notes & Findings

Automated multi-modal inspection conducted by DocGuard v1.0 on 2026-10-30 19:46:09 UTC. The primary neural fusion classifier assigned a 12.50% probability of tampering. Document exhibits complete physical and mathematical equilibrium across all analyzed forensic vectors.

LEGAL EVIDENTIARY CERTIFICATION (FRE 902(13)/(14) & ISO/IEC 27037):
This certificate was generated by an automated, self-authenticating digital forensics pipeline. The cryptographic hash computed at acquisition establishes that the target document has not undergone unauthorized alteration since timestamp capture. All algorithms, feature extractors, and weights are version-controlled and reproducible. This certificate is suitable for submission in civil, regulatory, and insurance dispute proceedings.

DIGITAL SEAL
DOCGUARD
VERIFIED AUDIT